

SQL PRACTICE WORKBOOK

Questions by **Instructor Ahmed Salem**

SQL Practice Questions

*60 practice questions across seven progressive levels —
from a simple SELECT to compound analytical queries*

SELECT & WHERE

ORDER BY & LIMIT

JOIN

Aggregate Functions

GROUP BY

HAVING

Case Study

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Seven levels of increasing difficulty, each building on the one before it. Numbers on the right refer to the question numbering used throughout the booklet.

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01

Level One

SELECT & WHERE · Basic queries and filtering results with conditions

- 01 Show all the products in the `products` table.
- 02 Show only the product names and sale prices.
- 03 Show the products whose sale price is greater than 30.
- 04 Show the products whose purchase price is less than 20.
- 05 Show the products whose sale price is between 20 and 50.
- 06 Show the products where `is_active = true`.
- 07 Show all suppliers with a `balance` greater than 0.
- 08 Show the customers who have no outstanding balance.
- 09 Show the products whose minimum stock level (`stock_min`) is greater than 20.
- 10 Show the products whose name contains the word "Milk".

02

Level Two

ORDER BY & LIMIT · Sorting results and limiting how many rows are returned

11 Show the products sorted by sale price from lowest to highest.

12 Show the products sorted by sale price from highest to lowest.

13 Show the 5 most expensive products.

14 Show the 5 cheapest products.

15 Show the last 5 sales invoices by date.

16 Show the suppliers sorted by balance from highest to lowest.

03

Level Three

JOIN · Combining data across related tables

17 Show each product name together with its category name.

REQUIRED OUTPUT:

Product Name

Category Name

18 Show all the products that belong to the “Drinks” category.

19 Show all the products that belong to the “Dairy” category.

20 Show the purchase invoices together with the supplier name.

REQUIRED OUTPUT:

Invoice Number

Supplier Name

Invoice Date

Total Amount

21 Show the sales invoices together with the customer name.

22 Show all sales invoices, including invoices that have no customer.

23 Show the sales invoice details so that the output includes:

Invoice Number

Product Name

Quantity

Sale Price

Total

24 Show the purchase invoice details so that the output includes:

Invoice Number
Supplier Name
Product Name
Quantity
Purchase Price
Total

04

Level Four

Aggregate Functions · Core aggregate functions

- | | |
|----|---|
| 25 | Count the number of products in the database. |
| 26 | Calculate the average sale price across all products. |
| 27 | Find the highest sale price. |
| 28 | Find the lowest sale price. |
| 29 | Calculate the total value of all sales invoices. |
| 30 | Calculate the total value of all purchase invoices. |
| 31 | Calculate the total amount paid on sales invoices. |
| 32 | Calculate the total amount still owed by customers. |
| 33 | Calculate the total amount still owed to suppliers. |

05

Level Five

GROUP BY · Grouping results by a category or entity

34 Count the number of products in each category.

REQUIRED RESULT:

Category **Name**

Products Count

35 Calculate the average sale price for each category.

36 Find the highest sale price in each category.

37 Calculate the total sales for each customer.

38 Count the number of sales invoices for each customer.

39 Calculate the total purchases from each supplier.

40 Count the number of purchase invoices for each supplier.

41 Calculate the total quantity sold for each product.

42 Show the top 5 best-selling products by quantity.

06

Level Six

HAVING · Filtering the results of a GROUP BY

- 43 Show the categories that contain more than 3 products.
- 44 Show the customers whose total purchases exceed 200.
- 45 Show the suppliers whose total purchase value exceeds 2000.
- 46 Show the products that have sold more than 5 units.
- 47 Show the categories whose average sale price is greater than 30.

07

Level Seven

Case Study · Applied questions that combine everything above

- 48 Which product has the highest sale price?
- 49 Which product has the lowest sale price?
- 50 Which product has sold the most by quantity?
- 51 Which category contains the largest number of products?
- 52 Which customer has the highest total purchases?
- 53 Which supplier accounts for the largest purchase value?
- 54 What is the total value of sales on 2026-09-14?
- 55 What is the total value of sales between 2026-09-10 and 2026-09-14?
- 56 Show the products that have been sold but whose current sale price differs from the price recorded in `items_sale`.
- 57 Show all invoices that still have an outstanding balance.
- 58 Show all sales invoices that have not been paid in full.
- 59 Show the products that have been purchased but never sold.

60 Show the products that have been sold but never appear on any purchase invoice.